How Literacy and Age at Marriage Shape Family Size in Portugal, 1979*

A Statistic Research Using Poisson (Negative Binominal) Regression Models

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TBD. IDK. Smash Them

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*Code and data are available at: https://github.com/Jie-jiao05/Family_Size.

1 Introduction

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3 Result

3.1 Data Description

This dataset originates from the 1979-80 Portuguese Fertility Survey (Instituto Nacional de Estatística (Portugal) 1979), conducted as part of the World Fertility Survey (WFS) program and archived by Princeton University's Office of Population Research (OPR).

The Dataset includes 5,148 observations and 8 variables, covering factors such as literacy, marriage age, pregnancies, children, and regional demographics. Table 1 it shows a statistical summary of our dataset.

To analyze the relationship between family size, literacy, and age at marriage, the variable Literacy was categorized as a dummy variable, assigning values 0 and 1 to illiterate and literate individuals, respectively. The analysis primarily focused on demographic factors, economic influences, and regional differences, aiming to assess their impact on family size. Additionally, an offset variable accounting for the duration of marriage was included to control for varying exposure times.

Table 1: Dataset Summary

age	ageMarried	monthsSinceM	pregnancies
Min. :15.00	0to15 : 52	Min. : 0.0	Min. : 0.000
1st Qu.:28.75	15to18 : 452	1st Qu.: 73.0	1st Qu.: 1.000
Median :36.00	18to20 : 910	Median :148.0	Median : 2.000
Mean :35.30	20to22 :1126	Mean :155.2	Mean : 2.282
3rd Qu.:43.00	22to25 :1468	3rd Qu.:230.0	3rd Qu.: 3.000
Max. :49.00	25to30 : 923	Max. :431.0	Max. :16.000
	30toInf: 217		

children	sons	region	literacy
Min. : 0.00	Min. : 0.000	10-20k: 433	no : 581
1st Qu.: 1.00	1st Qu.: 0.000	20k+ : 583	yes:4567
Median : 2.00	Median : 1.000	lisbon: 470	
Mean : 2.26	Mean : 1.187	lt10k :3502	
3rd Qu.: 3.00	3rd Qu.: 2.000	porto : 160	
Max. :17.00	Max. :10.000		

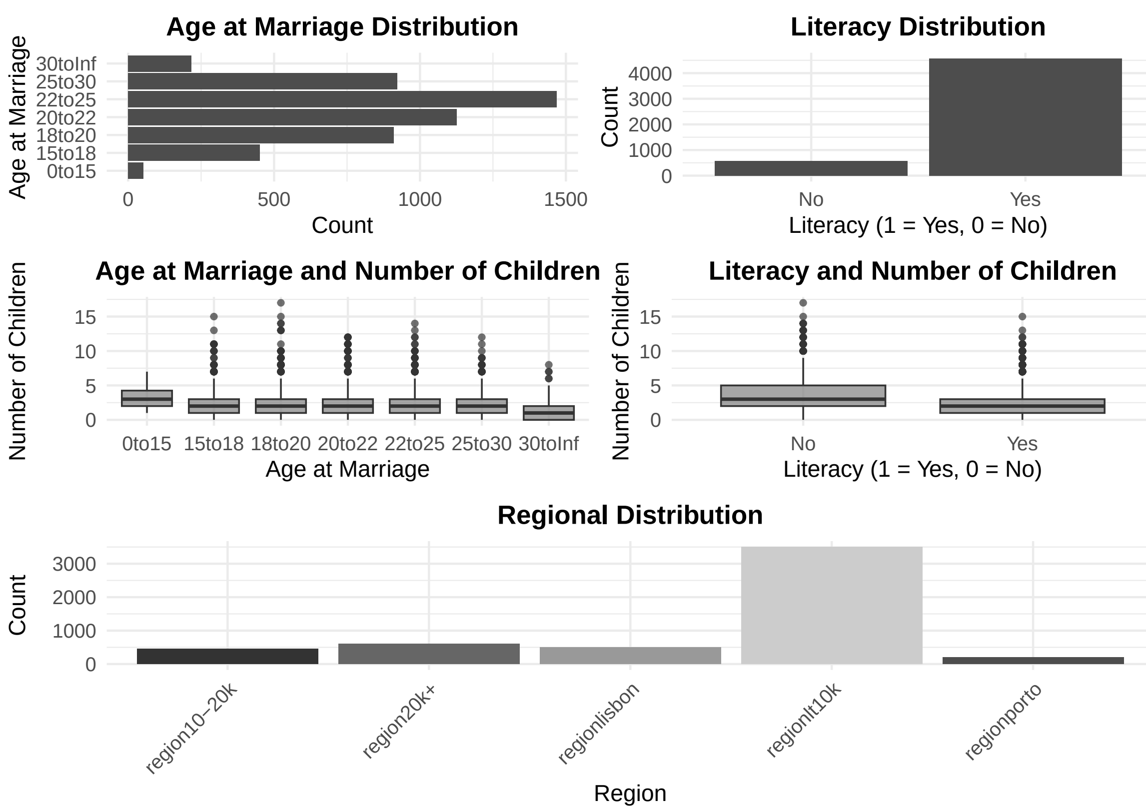

Figure 1: Distribution of Key Variable

Figure 1 illustrates the distribution of key variables and their relationship with family size. In general, the age of marriage is normally distributed within the dataset, slightly skewed to the right, with a peak at 22-25 years old. At the same time, the Portuguese have a low level of education, most of them come from the lt10k area, and illiterate people exist only as a minority in society. From the boxplot of the age of marriage and the number of children, we can see that the number of children is inversely proportional to the age. The lower the age of marriage, the more children there are. At the same time, those who are not educated generally have more children.

But unignorable, the vast majority of samples come from region_lt10k, creating a highly imbalanced data distribution, which may lead to increased variance and potential overdispersion. To account for differences in exposure across groups, we will consider an offset term during model selection, adjusting for variations in time since marriage. Details will be discussed in Section 3.2 and Section 4.1.

3.2 Model Selection

For model Selection we consider 3 model under following condition and conduct Log_Likelihood test, AIC, Deviance to make comparison cross model, and comes with our final model. The detailed models are list below. And for the consideration of offset and over-dispersion are shown in Section 4.1

Model 1: Poisson Regression with Offset

$$\begin{aligned}
\text{children}_i &\sim \text{Poisson}(\lambda_i) && (1)\\
\log(\lambda_i) &= \beta_0 + \beta_1 \cdot \text{literacy}_i && (2)\\
&\quad + \beta_2 \cdot \text{ageMarried}_{15-18,i} + \beta_3 \cdot \text{ageMarried}_{18-20,i} && (3)\\
&\quad + \beta_4 \cdot \text{ageMarried}_{22-25,i} + \beta_5 \cdot \text{ageMarried}_{25-30,i} && (4)\\
&\quad + \beta_6 \cdot \text{ageMarried}_{30+,i} + \beta_7 \cdot \text{region}_{10-20k,i} && (5)\\
&\quad + \beta_8 \cdot \text{region}_{20k+,i} + \beta_9 \cdot \text{region}_{Lisbon,i} && (6)\\
&\quad + \beta_{10} \cdot \text{region}_{lt10k,i} + \log(\text{monthsSinceM}_i) && (7)
\end{aligned}$$

Model 2: Negative Binomial Regression with offset

$$
\begin{aligned}
\text{children}_i &\sim \text{Negative Binomial}(\lambda_i, \theta) && (8)\\
\log(\lambda_i) &= \beta_0 + \beta_1 \cdot \text{literacy}_i && (9)\\
&+ \beta_2 \cdot \text{ageMarried}_{15-18,i} + \beta_3 \cdot \text{ageMarried}_{18-20,i} && (10)\\
&+ \beta_4 \cdot \text{ageMarried}_{22-25,i} + \beta_5 \cdot \text{ageMarried}_{25-30,i} && (11)\\
&+ \beta_6 \cdot \text{ageMarried}_{30+,i} + \beta_7 \cdot \text{region}_{10-20k,i} && (12)\\
&+ \beta_8 \cdot \text{region}_{20k+,i} + \beta_9 \cdot \text{region}_{Lisbon,i} && (13)\\
&+ \beta_{10} \cdot \text{region}_{lt10k,i} + \log(\text{monthsSinceM}_i) && (14)
\end{aligned}
$$

Model 3: Poisson Regression without Offset

$$
\begin{aligned}
\text{children}_i &\sim \text{Poisson}(\lambda_i) && (15)\\
\log(\lambda_i) &= \beta_0 + \beta_1 \cdot \text{literacy}_i && (16)\\
&+ \beta_2 \cdot \text{ageMarried}_{15-18,i} + \beta_3 \cdot \text{ageMarried}_{18-20,i} && (17)\\
&+ \beta_4 \cdot \text{ageMarried}_{22-25,i} + \beta_5 \cdot \text{ageMarried}_{25-30,i} && (18)\\
&+ \beta_6 \cdot \text{ageMarried}_{30+,i} + \beta_7 \cdot \text{region}_{10-20k,i} && (19)\\
&+ \beta_8 \cdot \text{region}_{20k+,i} + \beta_9 \cdot \text{region}_{Lisbon,i} && (20)\\
&+ \beta_{10} \cdot \text{region}_{lt10k,i} && (21)
\end{aligned}
$$

Table 2: Likelihood Ratio Test Results

Comparison	Log_Likelihood	Chi_Square	p_Value	AIC
Model 1	-8944.7	NA	NA	17913.42
Model 2	-8909.1	NA	NA	17842.12
Model 3	-9234.8	NA	NA	18493.57
Model 1 vs Model 2	NA	71.307	2.2e-16	NA
Model 1 vs Model 3	NA	580.150	2.2e-16	NA
Model 2 vs Model 3	NA	651.450	2.2e-16	NA

From Table 2, we observe that, based on the Log-Likelihood test, Model 2 provides the most accurate results as it effectively accounts for both overdispersion and the offset. Additionally, Model 2 has the lowest AIC (17,842.12) and deviance (4,912.803), indicating a better overall fit compared to Models 1 and 3. Therefore, we select Model 2 as our final model.

3.3 Final Model

Our final model:model 2 consider over-dispersion with offset is showing below, statistic summary are showing in Table 3

Final Model: Model 2: Negative Binomial Regression with offset

$$
\begin{aligned}
\text{children}_i &\sim \text{Negative Binomial}(\lambda_i, \theta) && (22)\\
\log(\lambda_i) &= -4.14332 - 0.11911 \cdot \text{literacy}_i && (23)\\
&+ 0.05177 \cdot \text{ageMarried}_{15-18,i} + 0.03431 \cdot \text{ageMarried}_{18-20,i} && (24)\\
&- 0.03097 \cdot \text{ageMarried}_{22-25,i} - 0.01469 \cdot \text{ageMarried}_{25-30,i} && (25)\\
&- 0.00269 \cdot \text{ageMarried}_{30+,i} - 0.064757 \cdot \text{region}_{10-20k,i} && (26)\\
&- 0.19794 \cdot \text{region}_{20k+,i} - 0.17982 \cdot \text{region}_{Lisbon,i} && (27)\\
&+ 0.09661 \cdot \text{region}_{lt10k,i} + \log(\text{monthsSinceM}_i) && (28)
\end{aligned}
$$

Reference Categories:

- **Age at Marriage:** 0-15 is the reference category .
- **Region:** Porto is the reference category .

Table 3: Final Model

	Estimate	Std. Error	z value	Pr(>|z|)	Rate Ratio	2.5%	97.5%
(Intercept)	-4.143	0.064	-64.997	0.000	0.016	0.014	0.018
literacy	-0.119	0.026	-4.514	0.000	0.888	0.843	0.935
ageMarried_15_18	0.052	0.038	1.357	0.175	1.053	0.977	1.135
ageMarried_18_20	0.034	0.031	1.096	0.273	1.035	0.973	1.100
ageMarried_22_25	-0.031	0.028	-1.094	0.274	0.970	0.917	1.025
ageMarried_25_30	-0.015	0.032	-0.458	0.647	0.985	0.925	1.049
ageMarried_30_plus	0.003	0.063	-0.043	0.966	0.997	0.882	1.128
region_10_20k	-0.065	0.067	-0.960	0.337	0.937	0.821	1.070
region_20k_plus	-0.198	0.066	-3.018	0.003	0.820	0.721	0.933
region_lisbon	-0.180	0.068	-2.639	0.008	0.835	0.731	0.955
region_lt10k	0.097	0.058	1.665	0.096	1.101	0.983	1.234

3.4 Anova Test of Region, Marry Age and Literacy

Our research aim to analysis how marry age and literacy will have impact to the size of family cross region, to test that, we conduct anova test. The result are shown in Table 4, from the data we can see all p-value are less than 0.05 so we fail to reject our null hypothesis, which indicate literacy and region are significant to our model.

Table 4: Anova Test Result

Comparison	Chi_Square	p_Value	df
Full Model vs No Literacy	19.905	8.137e-06	1
Full Model vs No Region	118.890	2.2e-16	4
Full Model vs No Marriage Age	748.720	2.2e-16	4

4 Conclusion

Our final model focus that both literacy and age at marriage are significant to family size across region , with age at marriage having the most substantial impact.

Using marriage age 0–15 and Porto as reference categories, literate individuals have 11.2% fewer children than illiterate individuals, suggesting that lower education levels are associated with larger family sizes. Additionally, regional disparities are evident, with families in Lisbon having a family size that is 0.835 times that of those in Porto, indicating that less developed regions tend to have larger families. Compare to Lisbon and Porto, a literate individual in Lisbon married between 22–25 is expected to have 25.9% fewer children than an illiterate counterpart in Porto.

However, age at marriage does not significantly affect family size, as all p-values for marriage age categories are above 0.05. Suggesting that variations in marriage timing alone do not drive differences in fertility when controlling for literacy and regional factors.

Comparing Lisbon and Porto, a literate individual in Lisbon married between 22–25 is expected to have 25.9% fewer children than an illiterate counterpart in Porto, reinforcing the combined impact of education, regional context, and marital timing on fertility outcomes.

Appendix

4.1 Over-Dispersion and Offset

our coefficient plot Figure 2 shows that adding an offset (log(monthsSinceM)) reduces the magnitude of estimates (from blue to orange), help to stabilize variance. And Transform into Negative Binomial distribution(green) further refines confidence intervals while preserving coefficient. And the dispersion ratio is 1.24, this suggests a mild overdispersion, making the Negative Binomial model with an offset the a better choice.

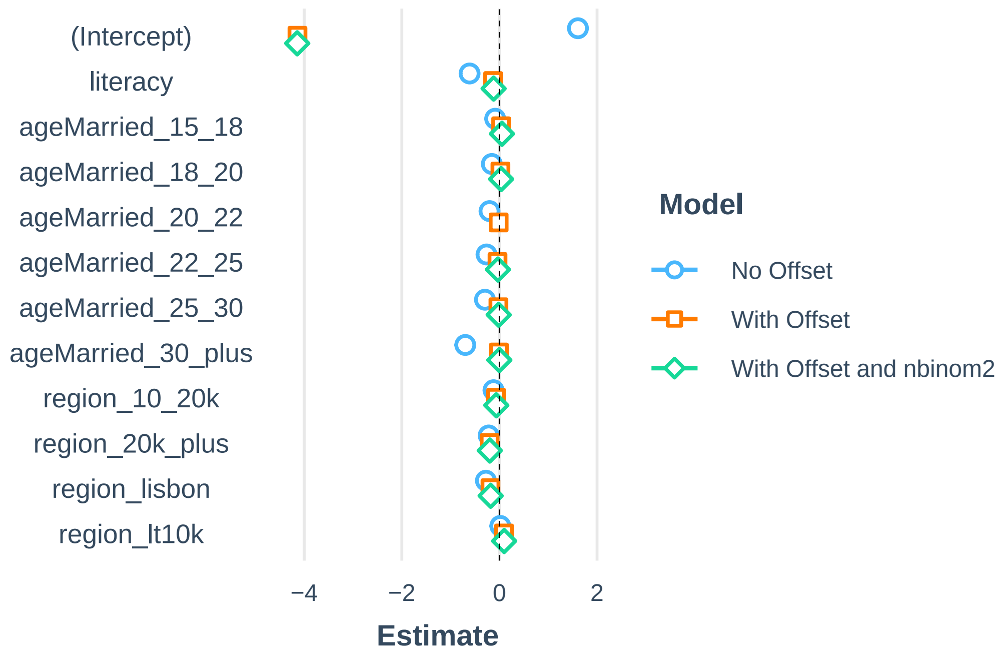

Figure 2: Distribution of Key Variable

References

Instituto Nacional de Estatística (Portugal). 1979. *Portugal World Fertility Survey (WFS), 1979-80*. World Fertility Survey (WFS), Demographic; Health Surveys (DHS) Program. https://wfs.dhsprogram.com/index.cfm?ccode=pt.